

- No. of Printed Pages : 4
Roll No.

Time : 3Hrs. M.M. : 100

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- ### SECTION-D
- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain in detail the manufacturing process of Ferrites.
- Q.37 Write a short note on phenomenon and properties of superconductors.
- Q.38 Enlist modern ceramics products? Classify and explain with examples.

- Q.6 Soft ferrites are generally used in:
 a) Electronic Ballast b) Thermometers
 c) Optical fibers d) Bricks
- Q.7 A capacitor stores:
 a) Electric Charge b) Heat
 c) Light d) Current
- Q.8 Superconductors are:
 a) Paramagnetic b) Ferromagnetic
 c) Diamagnetic d) Anti-ferromagnetic
- Q.9 The phenomenon of generating electricity under mechanical stress is:
 a) Thermoelectric b) Pyroelectric
 c) Piezoelectric d) Magnetic
- Q.10 Dental ceramics are used for:
 a) Bone repair b) Heart surgery
 c) Tooth Capping d) Optical lens

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Ferrites have high electrical conductivity. (True/False)
- Q.12 Fuel rods in nuclear reactor are used to start the reaction. (True/False)
- Q.13 Varistors are used to protect against high temperature. (True/False)

- Q.14 Ceramic tiles are examples of advanced ceramics. (True/False)
- Q.15 Pyroelectric materials respond to temperature changes. (True/False)
- Q.16 Thermistor is a voltage-sensitive device. (True/False)
- Q.17 Bio ceramics help in healing bone fractures. (True/False)
- Q.18 PLZT is a ferroelectric material. (True/False)
- Q.19 Resistance of superconductors is zero. (True/False)
- Q.20 A superconductor behaves like a normal conductor above its critical temperature (True/False)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 List the classification of modern ceramics.
- Q.22 Explain the working and structure of multilayer capacitors.
- Q.23 Define Meissner effect and its relevance.
- Q.24 Describe the manufacture of dental ceramics.
- Q.25 Compare soft and hard ferrites.
- Q.26 Define dielectric material? Give examples.
- Q.27 Write a short note on development of high temperature superconductors.
- Q.28 Define electro-optic ceramics and give example.